

Name:

Roll:

(CG501) Computation and Cognition - Quiz 3

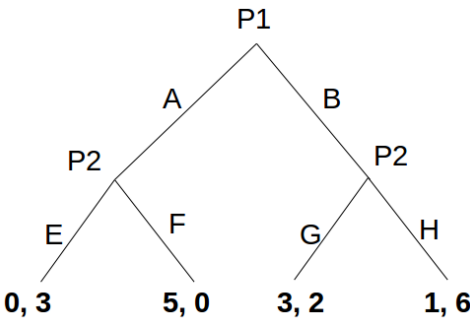
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1. Find the pure strategy Nash equilibrium in the static game given below:

	A	B
C	15, 10	10, 15
D	5, 11	11, 17

1 point

2. Find the Nash equilibrium in the given sequential game by using the method of backwards induction:



3 points

3. You and a friend are at a restaurant, and the owner offers you a free pizza under the following conditions. Both of you must simultaneously announce how many pieces you would like from the choices {2, 4, 6}. If the total number of pieces exceeds six, neither of you get any pizza. Otherwise, both of you get the number of pieces you asked. Assume that both of you care only about how much pizza you individually consume, and more is better. Write the payoff matrix of the static game. Find all pure strategy Nash equilibria. (Hint: be sure to check all cells)

4 points

4. Suppose you are a warlord in Palaj in medieval times, and you own two dragons. You’ve arrived at the following deal with slave masters of Chiloda: half the Fearless slave army for one dragon, or both the dragons for the entire Fearless army.

Assume that two dragons together are more powerful as weapons than single (i.e, their combined strength is more than the sum of individual strengths); similarly, the Fearless also wield less than 50 percent power when broken up into half. Assume that your payoff and the slave masters’ payoff are strictly given by the total military power available. Model the situation as a static game of complete information (no need to find equilibria), and explain each payoff pair. (Hint: You have two actions available: {sell one dragon, sell two dragons} and slave masters have two actions available: {sell half army, sell full army})

4 points